OTZ - Optical Trapping

Signature Sheet

Student's Name _____ Partner's Name _____

Pre-Lab Discussion Questions

It is your responsibility to discuss this lab with an instructor before your first day of your scheduled lab period. This signed sheet must be included as the first page of your report. Without it you will lose grade points. You should be prepared to discuss at least the following before you come to lab:

1. How does an Optical Trap work? Show us the optical path of the trapping laser. What is the trapping laser's wavelength?
2. What is a Power Spectrum Density (PSD) Graph and what can it be used for?
3. What is meant by sensitivity and stiffness of an Optical Trap?
4. What are the safety requirements for working with this laser?

Staff Signature _____ Date _____

Completed before the first day of lab? (Circle one) Yes / No

Mid-Lab Discussion Questions

1. By day 3 of this lab, you should have successfully created a slide with a dilute solution of 1 micron beads, turned on the laser, trapped a bead, and moved it vertically and horizontally. Start the Optical Trapping program and take real time data while the bead is trapped. Using the "Alt + Print Screen" command, copy the Power Spectrum Density graph and show it to a GSI.

Staff Signature _____ Date _____

Completed by day 3 of lab? (Circle one) Yes / No

Checkpoint Signatures

1. Laser Goggle Safety Check

   Staff Signature _____

2. Trapping a Single Bead

   Staff Signature _____

3. Sensitivity vs Laser Power Plot

   Staff Signature _____

4. Sensitivities From Alpha Values

   Staff Signature _____

5. Stiffness Values

   Staff Signature _____

6. Two Sets of Questions

   Staff Signature _____

Station Clean-Up Check-Off

Before leaving, ensure your station is clean and ready for the next group. The GSI must verify and sign off on the following:

1. Any BNC cables that you have been moving are unplugged; return things to the state that you found them.
2. AFM prepared samples are removed and the sample disks are returned to the bag. CD/DVD/Blu-Ray scraps are thrown out if small, put in the bin if large.
3. All used pipette tips go into trash and glass sample slides go into glass disposal container; pipettes are put back to the shelf.
4. Workspace cleared of personal belongings, papers, and debris.
5. Any spills or hazards cleaned up; no left-over onion in the fridge or on the bench.
6. Microscope is clear of mounted sample, turned off and covered by the plastic bag properly if applicable.
7. Restart the computer at your station so that any software updates are applied.
8. Station is clean and ready for the next group. Staff discretion for additional cleanup that is necessary applies.

Staff Signature _____ Date _____